

1. COMMON CODE — WITHOUT SESSION

HTML Form

```
<form action="CommonServlet" method="post">

<input type="text" name="field1" placeholder="Field1"><br><br>

<input type="text" name="field2" placeholder="Field2"><br><br>

<input type="text" name="field3" placeholder="Field3"><br><br>

<input type="submit" value="Submit">

</form>
```

CommonServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class CommonServlet extends HttpServlet
{
    public void doPost(HttpServletRequest req,
                        HttpServletResponse res)
                        throws IOException
    {
        res.setContentType("text/html");

        PrintWriter out = res.getWriter();

        // Get values
        String field1 = req.getParameter("field1");
        String field2 = req.getParameter("field2");
        String field3 = req.getParameter("field3");

        boolean valid = true;

        // Validation
        if(field1.equals(""))
        {
            out.println("Field1 Required<br>");
            valid = false;
        }

        if(field2.equals(""))
        {
            out.println("Field2 Required<br>");
            valid = false;
        }

        if(field3.equals(""))
```

```
        {
            out.println("Field3 Required<br>");
            valid = false;
        }

        // Success
        if(valid)
        {
            out.println("Form Submitted Successfully");
        }
    }
}
```

2. COMMON CODE — WITH SESSION

HTML Form

```
<form action="SessionServlet" method="post">

<input type="text" name="field1"><br><br>

<input type="text" name="field2"><br><br>

<input type="submit" value="Submit">

</form>
```

SessionServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class SessionServlet extends HttpServlet
{
    public void doPost(HttpServletRequest req,
                        HttpServletResponse res)
                        throws IOException
    {
        res.setContentType("text/html");

        PrintWriter out = res.getWriter();

        String field1 = req.getParameter("field1");
        String field2 = req.getParameter("field2");

        // Session object
        HttpSession session = req.getSession();

        // Store values
        session.setAttribute("f1", field1);
        session.setAttribute("f2", field2);

        out.println("Stored Successfully<br>");
    }
}
```

```
        out.println(session.getAttribute("f1")+"<br>");  
        out.println(session.getAttribute("f2"));  
    }  
}
```

WHAT TO CHANGE FOR EACH QUESTION

1. USER REGISTRATION FORM

Fields

```
username  
email  
password  
confirm password
```

Replace in HTML

```
<input type="text" name="username">  
  
<input type="text" name="email">  
  
<input type="password" name="password">  
  
<input type="password" name="cpassword">
```

Replace in Servlet

```
String username = req.getParameter("username");  
  
String email = req.getParameter("email");  
  
String password = req.getParameter("password");  
  
String cpassword = req.getParameter("cpassword");
```

Validation

```
if(!email.contains("@"))  
  
if(password.length() < 6)  
  
if(!password.equals(cpassword))
```

2. E-COMMERCE CHECKOUT

Fields

```
name  
address  
postalcode  
phone  
cardnumber  
expirydate
```

Replace in Servlet

```
String postal = req.getParameter("postalcode");  
  
String phone = req.getParameter("phone");  
  
String card = req.getParameter("cardnumber");  
  
String expiry = req.getParameter("expirydate");
```

Validation

```
if(postal.length() != 6)  
  
if(phone.length() != 10)  
  
if(card.length() != 16)
```

3. JOB APPLICATION

Fields

```
name  
qualification  
experience  
resume
```

Replace in Servlet

```
String name = req.getParameter("name");  
  
String qualification = req.getParameter("qualification");  
  
String experience = req.getParameter("experience");
```

Validation

```
if (name.equals(""))  
  
if (qualification.equals(""))  
  
if (experience.equals(""))
```

4. EVENT REGISTRATION

Fields

```
name  
email  
age  
tickets
```

Replace in Servlet

```
String age = req.getParameter("age");  
  
String tickets = req.getParameter("tickets");
```

Validation

```
int a = Integer.parseInt(age);  
  
if (a < 18)  
  
if (email.equals("admin@gmail.com"))
```

Cost Calculation

```
int t = Integer.parseInt(tickets);  
  
int total = t * 500;
```

5. SURVEY FORM

Fields

```
name  
feedback  
rating
```

Replace in Servlet

```
String feedback = req.getParameter("feedback");  
String rating = req.getParameter("rating");
```

Validation

```
if(feedback.equals(""))  
  
if(rating.equals(""))
```

1. MAIN PAGE (index.html)

```
<html>  
  
<body>  
  
<h2>Image Map Example</h2>  
  
  
  
<map name="mymap">  
  
<!-- Area 1 -->  
<area shape="rect"  
      coords="50,50,150,150"  
      href="area1.html">  
  
<!-- Area 2 -->  
<area shape="rect"  
      coords="200,50,300,150"  
      href="area2.html">  
  
<!-- Area 3 -->  
<area shape="rect"  
      coords="350,50,450,150"  
      href="area3.html">  
  
</map>  
  
</body>  
  
</html>
```

1.GEOGRAPHIC INFORMATION SYSTEM

Areas

- Tamil Nadu
- Kerala
- Karnataka

area1.html — Tamil Nadu

```
<html>
<body>

<h2>Tamil Nadu</h2>

<ul>
<li>Population : 7 Crore</li>
<li>Capital : Chennai</li>
<li>Main Industry : Textile</li>
</ul>

</body>
</html>
```

area2.html — Kerala

```
<html>
<body>

<h2>Kerala</h2>

<ul>
<li>Population : 3 Crore</li>
<li>Capital : Thiruvananthapuram</li>
<li>Main Industry : Tourism</li>
</ul>

</body>
</html>
```

area3.html — Karnataka

```
<html>
<body>

<h2>Karnataka</h2>

<ul>
<li>Population : 6 Crore</li>
<li>Capital : Bengaluru</li>
```

```
<li>Main Industry : IT</li>
</ul>
```

```
</body>
</html>
```

2. INTERACTIVE WORLD MAP

Areas

- Asia
 - Europe
 - Africa
-

area1.html — Asia

```
<html>
<body>

<h2>Asia</h2>

<ul>
<li>Population : 4.7 Billion</li>
<li>Major City : Tokyo</li>
<li>Largest Continent</li>
</ul>

</body>
</html>
```

area2.html — Europe

```
<html>
<body>

<h2>Europe</h2>

<ul>
<li>Population : 750 Million</li>
<li>Major City : London</li>
<li>Famous for Industries</li>
</ul>

</body>
</html>
```

area3.html — Africa


```
<html>
<body>

<h2>Africa</h2>

<ul>
<li>Population : 1.4 Billion</li>
<li>Major City : Cairo</li>
<li>Rich in Minerals</li>
</ul>

</body>
</html>
```

3. SOLAR SYSTEM MAP

Areas

- Earth
 - Mars
 - Jupiter
-

area1.html — Earth

```
<html>
<body>

<h2>Earth</h2>

<ul>
<li>Third Planet</li>
<li>Supports Life</li>
<li>One Moon</li>
</ul>

</body>
</html>
```

area2.html — Mars

```
<html>
<body>

<h2>Mars</h2>

<ul>
<li>Known as Red Planet</li>
<li>Two Moons</li>
<li>NASA Missions Sent</li>
</ul>
```

```
</body>
</html>
```

area3.html — Jupiter

```
<html>
<body>

<h2>Jupiter</h2>

<ul>
<li>Largest Planet</li>
<li>Many Moons</li>
<li>Gas Giant</li>
</ul>

</body>
</html>
```

4. COMPUTER SYSTEM MAP

Areas

- CPU
 - RAM
 - Motherboard
-

area1.html — CPU

```
<html>
<body>

<h2>CPU</h2>

<ul>
<li>Main Processor</li>
<li>Performs Calculations</li>
<li>Controls Computer</li>
</ul>

</body>
</html>
```

area2.html — RAM

```
<html>
```

```
<body>

<h2>RAM</h2>

<ul>
<li>Temporary Memory</li>
<li>Improves Speed</li>
<li>Stores Running Data</li>
</ul>

</body>
</html>
```

area3.html — Motherboard

```
<html>
<body>

<h2>Motherboard</h2>

<ul>
<li>Main Circuit Board</li>
<li>Connects Components</li>
<li>Supports Hardware</li>
</ul>

</body>
</html>
```

5. COLLEGE CAMPUS MAP

Areas

- CSE Department
 - Library
 - Office
-

area1.html — CSE Department

```
<html>
<body>

<h2>CSE Department</h2>

<ul>
<li>Computer Labs Available</li>
<li>Experienced Staff</li>
<li>Placement Training</li>
</ul>
```

```
</body>
</html>
```

area2.html — Library

```
<html>
<body>

<h2>Library</h2>

<ul>
<li>Thousands of Books</li>
<li>Digital Section</li>
<li>Reading Hall</li>
</ul>

</body>
</html>
```

area3.html — Office

```
<html>
<body>

<h2>Office</h2>

<ul>
<li>Student Records</li>
<li>Fee Payment</li>
<li>Certificate Issue</li>
</ul>

</body>
</html>
```

COMMON DHTML TEMPLATE

```
<html>
<head>
<title>Common Form</title>

<script src="validation.js"></script>

</head>

<body>

<h2>Dynamic Form Validation</h2>
```

```
<form onsubmit="return validateForm()">

<!-- FIELD 1 -->
<label>Field 1</label>
<input type="text" id="field1">
<span id="msg1"></span>
<br><br>

<!-- FIELD 2 -->
<label>Field 2</label>
<input type="text" id="field2">
<span id="msg2"></span>
<br><br>

<!-- FIELD 3 -->
<label>Field 3</label>
<input type="text" id="field3">
<span id="msg3"></span>
<br><br>

<!-- SUBMIT -->
<input type="submit" id="submitBtn" value="Submit">

</form>

</body>
</html>
```

COMMON VALIDATION SCRIPT (ALL- IN-ONE FILE)

```
function validateRegistration()
{
    let username = document.getElementById("username").value;
    let email = document.getElementById("email").value;
    let password = document.getElementById("password").value;
    let cpassword = document.getElementById("cpassword").value;

    let ok = true;

    if(username.length < 3)
    {
        ok = false;
        alert("Username too short");
    }

    if(!email.includes("@"))
    {
        ok = false;
        alert("Invalid email");
    }

    if(password.length < 8 ||
        !/[A-Z]/.test(password) ||
```

```
        !/[0-9]/.test(password) ||
        !/[!@#$$%^&*]/.test(password))
    {
        ok = false;
        alert("Weak password");
    }

    if(password != cpassword)
    {
        ok = false;
        alert("Password mismatch");
    }

    return ok;
}
```

```
function validateCheckout()
{
    let postal = document.getElementById("postal").value;
    let phone = document.getElementById("phone").value;
    let card = document.getElementById("card").value;
    let expiry = document.getElementById("expiry").value;

    let ok = true;

    if(postal.length != 6)
    {
        ok = false;
        alert("Invalid postal code");
    }

    if(phone.length != 10)
    {
        ok = false;
        alert("Invalid phone number");
    }

    if(card.length != 16)
    {
        ok = false;
        alert("Invalid card number");
    }

    // expiry check
    let today = new Date();
    let exp = new Date(expiry);

    if(exp < today)
    {
        ok = false;
        alert("Card expired");
    }

    return ok;
}
```

```
function validateJob()
{
    let name = document.getElementById("name").value;
    let start = document.getElementById("start").value;
    let end = document.getElementById("end").value;
```

```
    let file = document.getElementById("resume").value;

    let ok = true;

    if(name == "")
    {
        ok = false;
        alert("Name required");
    }

    if(new Date(start) > new Date(end))
    {
        ok = false;
        alert("Invalid date order");
    }

    if(!file.endsWith(".pdf"))
    {
        ok = false;
        alert("Only PDF allowed");
    }

    return ok;
}

function validateEvent()
{
    let email = document.getElementById("email").value;
    let age = document.getElementById("age").value;
    let session = document.getElementById("session").value;

    let ok = true;

    if(!email.includes("@"))
    {
        ok = false;
        alert("Invalid email");
    }

    if(age < 18)
    {
        ok = false;
        alert("Age restriction not met");
    }

    if(session == "")
    {
        ok = false;
        alert("Select session");
    }

    return ok;
}

function validateSurvey()
{
    let q1 = document.getElementById("q1").value;
    let rating = document.getElementById("rating").value;
    let comment = document.getElementById("comment").value;

    let ok = true;
```

```
    if(q1 == "")
    {
        ok = false;
        alert("Answer required");
    }

    if(rating < 1 || rating > 5)
    {
        ok = false;
        alert("Rating must be 1 to 5");
    }

    if(comment.length > 200)
    {
        ok = false;
        alert("Comment too long");
    }

    return ok;
}
```